get_wifi

Command: `get_wifi`

Description

Retrieves the **current Wi-Fi configuration** of the scanner — connected network, IP address, signal, and security settings.

Command details:

- Pattern Name: Wi-Fi Configuration Retrieval
- Communication Type: Bidirectional (Cloud to Device, Device to Cloud)
- MQTT Version: 3.1.1
- API Version: V1.1
- Document Version: 1.0.0
- Last Updated: 2026-03-12
- Applies To: RFD40 Series, RFD90 Series

Response fields:

Field	What it tells you
`interfaceName`	Wi-Fi interface identifier (e.g. `wlan0`)
`status`	Whether the Wi-Fi interface is Enabled or Disabled
`hostname`	The scanner's network hostname
`macAddress`	Wi-Fi MAC address
`accessPoint.essid`	The SSID of the connected Wi-Fi network
`accessPoint.status`	Connection state — Connected or Disconnected
`accessPoint.securityType`	Security protocol — WPA2Personal, WPA2Enterprise, WPA3Personal, Open
`accessPoint.isPreferred`	Whether this is the preferred network
`accessPoint.autoConn`	Whether the scanner auto-connects to this network
`ipv4Configuration.ipAddress`	Current IP address assigned to the scanner
`ipv4Configuration.subnetMask`	Subnet mask
`ipv4Configuration.gateway`	Default gateway
`ipv4Configuration.enableDhcp`	Whether DHCP is enabled or using static IP
`ipv4Configuration.dnsServer`	DNS server address

Typical use cases:

- Network troubleshooting — verify the scanner is connected to the correct SSID with a valid IP.
- Fleet monitoring — check Wi-Fi status across all scanners from a central dashboard.
- Security audit — confirm scanners are using WPA2/WPA3 Enterprise, not open networks.

MQTT topics for this command:

- Command topic (Cloud to Device):

`<tenantId>/CTRL/clients/cmnd/<deviceSerial>`

- Response topic (Device to Cloud):

`<tenantId>/CTRL/clients/resp/<deviceSerial>`

Example:

- Command: `zebra/CTRL/clients/cmnd/RFD40-12345678`
- Response: `zebra/CTRL/clients/resp/RFD40-12345678`

Ensure these topic paths match your configured endpoint topics.

Download pdf: [Download get_wifi as PDF](https://aa5123.github.io/zebra-rfd40-rfd90-iot-docs/command-pdfs/get_wifi.pdf)

Request

Request Fields

Field	Type	Required	Description	Example
`command`	string	Yes	Command issued to get all the wifi profiles saved in device	`get_wifi`
`requestId`	string	Yes	A unique identifier for the request, allowing tracking and debugging of the operation.	`abc123`

Request Examples

Generated Example

```
{
  "command": "get_wifi",
  "requestId": "abc123"
}
```

Response

Response Fields

Field	Type	Required	Description	Example
`command`	string	Yes		`get_wifi`
`requestId`	string	Yes		`abc123`
`apiVersion`	string (enum)	Yes		`V1.1`
`wifiProfiles`	object	No		
`wifiProfiles.wifiConfig`	array	No		
`response`	object	Yes	Status of the command execution — check `code` to confirm success (0) or identify the error.	

Field	Type	Required	Description	Example
`response.code`	integer	Yes	Response code indicating success or failure. • **0** — Success • **1** — Command payload accepted • **2** — Invalid payload • **3** — Not able to retrieve information • **4** — Firmware update in progress • **5** — Can't reboot, inventory in progress • **6** — Region is not configured • **7** — Interface is not available • **8** — Insufficient flash size • **9** — File not found • **10** — Configuration already exists • **11** — Inventory in progress • **12** — No radio operation in progress • **13** — Firmware update failed • **14** — Battery too low for firmware update • **15** — WIFI: SSID not found • **16** — WIFI: Cannot delete active SSID • **17** — WIFI: SSID missing	`0`

Field	Type	Required	Description	Example
			• **18** — WIFI: SSID already exists • **19** — WIFI: SSID count overflow • **20** — WiFi is not supported • **21** — Certificate not found • **22** — Advanced configuration not set • **23** — Invalid enum value • **24** — Max 32 prefilters limit exceeded • **25** — Max 3 publish topics exceeded • **26** — Max 1 subscribe topic exceeded • **27** — Invalid tenant ID length • **28** — Tag match pattern length exceeded	
`response.description`	string	Yes	Human-readable description of the response code (e.g. "Success", "Invalid payload").	`Success`

Response Examples

WPA2 Personal connection

```
{
  "command": "get_wifi",
  "requestId": "abc123",
  "apiVersion": "V1.1",
  "wifiConfig": {
    "interfaceDetails": {
      "interfaceName": "wlan0",
      "status": "Enabled",
      "hostname": "RFD40-22326520100477",
```

```
      "macAddress": "48:A4:93:BD:4A:FC",
      "accessPoint": {
        "essid": "IOT_TESTAP",
        "status": "Connected",
        "securityType": "WPA2Personal",
        "isPreferred": true,
        "autoConn": true
      },
      "ipv4Configuration": {
        "ipAddress": "192.168.0.104",
        "subnetMask": "255.255.255.0",
        "gateway": "192.168.0.1",
        "enableDhcp": true,
        "dnsServer": "1.0.168.192",
        "domainName": "zebra.com"
      }
    }
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```


WPA2 Enterprise connection


```
{
  "command": "get_wifi",
  "requestId": "abc002",
  "apiVersion": "V1.1",
  "wifiProfiles": {
    "wifiConfig": [
      {
        "interfaceDetails": {
          "interfaceName": "wlan0",
          "status": "ENABLED",
          "hostname": "RFD40-22326520100477",
          "macAddress": "E0-D0-45-3D-38-3F",
          "accessPoint": {
            "essid": "SECURE_AP",
            "status": "CONNECTED",
            "securityType": "WPA2Enterprise",
            "isPreferred": true,
            "autoConn": true
          },
          "ipv4Configuration": {
            "ipAddress": "172.16.10.88",
```

```
            "subnetMask": "255.255.255.0",
            "gateway": "172.16.10.1",
            "enableDhcp": true,
            "dnsServer": "172.16.10.5",
            "domainName": "secure.corp"
          },
          "securityDetails": {
            "WPA2Enterprise": {
              "authentication": "tls",
              "passphrase": "XXXXXXXX",
              "certificate": [
                {
                  "key": "ca_cert",
                  "value": "wifi_root_certificate"
                },
                {
                  "key": "client_cert",
                  "value": "wifi"
                },
                {
                  "key": "client_key",
                  "value": "wifi"
                }
              ],
              "protocol": "WPA2_Enterprise_CCMP"
            }
          }
        }
      }
    ]
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```

WPA3 Enterprise connection

```
{
  "command": "get_wifi",
  "requestId": "abc003",
  "apiVersion": "V1.1",
  "wifiProfiles": {
    "wifiConfig": [
      {
        "interfaceDetails": {
```

```
          "interfaceName": "wlan0",
          "status": "ENABLED",
          "hostname": "RFD40-22326520100477",
          "macAddress": "E0-D0-45-3D-38-4A",
          "accessPoint": {
            "essid": "WPA3_ENTERPRISE",
            "status": "CONNECTED",
            "securityType": "WPA3Enterprise",
            "isPreferred": false,
            "autoConn": false
          },
          "ipv4Configuration": {
            "ipAddress": "192.168.100.200",
            "subnetMask": "255.255.255.0",
            "gateway": "192.168.100.1",
            "enableDhcp": true,
            "dnsServer": "192.168.100.10",
            "domainName": "wpa3.enterprise.com"
          },
          "securityDetails": {
            "WPA3Enterprise": {
              "authentication": "ttls",
              "innerAuthentication": "mschapv2",
              "identity": "XXXXXXXX",
              "anonymousIdentity": "XXXXXXXX",
              "password": "XXXXXXXX",
              "certificate": [
                {
                  "key": "ca_cert",
                  "value": "wpa3_ca_cert"
                }
              ],
              "protocol": "WPA3_Enterprise_GCMP_256_SHA256"
            }
          }
        }
      }
    ]
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```


Open network connection

```
{
  "command": "get_wifi",
  "requestId": "open_network",
  "apiVersion": "V1.1",
  "wifiProfiles": {
    "wifiConfig": [
      {
        "interfaceDetails": {
          "interfaceName": "wlan0",
          "status": "ENABLED",
          "hostname": "RFD40-22326520100477",
          "macAddress": "E0-D0-45-3D-38-5B",
          "accessPoint": {
            "essid": "PUBLIC_WIFI",
            "status": "CONNECTED",
            "securityType": "Open",
            "isPreferred": false,
            "autoConn": false
          },
          "ipv4Configuration": {
            "ipAddress": "203.0.113.45",
            "subnetMask": "255.255.255.0",
            "gateway": "203.0.113.1",
            "enableDhcp": true,
            "dnsServer": "8.8.8.8",
            "domainName": "public.wifi"
          }
        }
      }
    ]
  },
  "response": {
    "code": 0,
    "description": "Success"
  }
}
```


WiFi interface disabled


```
{
  "command": "get_wifi",
  "requestId": "abc005",
  "apiVersion": "V1.1",
  "wifiConfig": {
    "interfaceDetails": {
      "interfaceName": "wlan0",
```

```
      "status": "DISABLED",
      "hostname": "RFD40-22326520100477",
      "macAddress": "E0-D0-45-3D-38-3F"
    }
  },
  "response": {
    "code": 7,
    "description": "Interface is not available"
  }
}
```